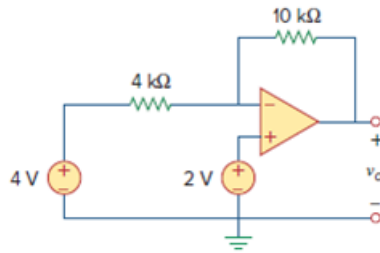


Problem

Calculate v_o in the op amp circuit of Fig. 5.60.

Figure 5. 60:



Step-by-step solution

Step 1 of 2

Refer to circuit diagram in Figure 5.60 in the textbook.

The non-inverting terminal is directly connected to 2 V voltage source.

The voltage at non-inverting terminal of the op-amp is,

$$v_+ = 2 \text{ V}$$

According to virtual ground concept, the voltages at inverting and non-inverting terminals of the op-amp are same for ideal op-amp.

$$\begin{aligned} v_- &= v_+ \\ &= 2 \text{ V} \end{aligned}$$

[Comments \(1\)](#)



Anonymous

Why are they both 2V though? I understand ideal means voltages are the same but what makes it 2V?

Step 2 of 2

Apply Kirchhoff's current law at inverting terminal of the op-amp.

$$\frac{v_- - 4 \text{ V}}{4 \text{ k}\Omega} + \frac{v_- - v_o}{10 \text{ k}\Omega} = 0$$

$$\frac{2 \text{ V} - 4 \text{ V}}{4 \text{ k}\Omega} + \frac{2 \text{ V} - v_o}{10 \text{ k}\Omega} = 0$$

$$\frac{v_o}{10 \text{ k}\Omega} = \frac{2 \text{ V}}{10 \text{ k}\Omega} - \frac{2 \text{ V}}{4 \text{ k}\Omega}$$

$$\begin{aligned} v_o &= \left(\frac{10 \text{ k}\Omega}{10 \text{ k}\Omega} \right) (2 \text{ V}) - \left(\frac{10 \text{ k}\Omega}{4 \text{ k}\Omega} \right) (2 \text{ V}) \\ &= 2 \text{ V} - (2.5)(2 \text{ V}) \\ &= 2 \text{ V} - 5 \text{ V} \\ &= -3 \text{ V} \end{aligned}$$

Therefore, the output voltage, v_o of op-amp is -3 V .